



WSSI Module – Complete Functional & Technical Guide

Objective

The **WSSI (Weekly Sales and Stock Intake)** module enables Format-Sales Channel-Brand-Segment-Fashion Grade level sales and inventory planning at a subperiod (week) level. It provides a comprehensive view of:

- Past performance (LY)
- Current year performance (CY) split into actuals and projected
- Baseline plans (AOP)

This document details the structure of the WSSI screen, all backend and frontend (UI) computed fields, and the logic behind every KPI. It is designed as a single reference for success teams, client teams, and technical users.



WSSI Screen Overview

The WSSI screen is structured into two major panels:

1. Sales Metrics Panel

Displays week-wise metrics under these sections:

- **Net Sales (Value)**
- **Unit Sales (Qty)**
- **Actual Selling Price (ASP)**
- **Discount (%)**
- **Cost of Goods Sold (COGS)** (*projected weeks only*)

Each metric has five rows:

- **CY (Actual/Projected)**
- **LY**
- **AOP**
- **Change over LY** (*UI calculated*)
- **Achievement against AOP** (*UI calculated*)

2. Inventory Metrics Panel

Includes:

- **Inventory Units** (Opening Stock, Target Inventory, LY, Change over LY)
- **Weeks of Cover** (Actual, Target, LY, Variance)
- **Inwards Units** (Planned Inwards, OTB, Additional Inwards)
- **Inventory COGS** (Opening)
- **Inwards COGS** (based on OTB)

- ◆ Note: Target Inventory, OTB, Weeks of Cover, Additional Inwards and all "Change over"/"Achievement" fields are calculated on UI.

In the columns, the subperiods are available which are segregated into the Actuals and



Inputs to the WSSI Module

The WSSI module depends on both **user-defined** and **system-integrated** inputs:

◆ User-Defined Inputs

1. Retail Week Calendar

Defines subperiods (weeks), periods (months), and the date ranges for forecast and LY analysis.

- `subperiod` , `period` , `forecast_start_date` , `forecast_end_date` , `analysis_start_date` , `analysis_end_date`

2. Performance Parameters

Defines AOP revenue, MRP, discount %, margin %, and cover durations across brand/channel/segment.

- 3 `aop_revenue` , `mrp` , `discount%` , `margin` , `cover_duration`
- Dimensions: Store, Channel, Brand, Segment, Fashion Grade, Period

3. Store Sales Channel Mapping

Maps each store to its primary sales channel and channel region.

- `store_code` , `channel` , `sales_channel`

4. Style Inventory Bucket

Maps styles to inventory buckets (X, Y, etc.).

- `style` , `inventory_bucket`

5. Inventory Bucket Flow

Defines inventory movement across sales channels.

- `inventory_bucket` , `flow` , `from_sales_channel` , `to_sales_channel` , `shelf_life` , `lead_time`

◆ System-Integrated Inputs

These are synced via ETL jobs or system integrations:

- **Store Sales** (`a_sales`)
- **Store Stock Snapshots** (`a_keyframe`)
- **Warehouse Stock** (`input_wssi_whstock`)
- **Store Inwards** (`input_wssi_store_inwards`)
- **Warehouse Inwards** (`input_wssi_wh_inwards`)
- **Open Orders, Goods in Transit (GIT)**

All data is synced only for the enabled stores, warehouses, brands, and categories, and only for the date range between the earliest `analysis_start_date` and the latest `forecast_end_date` in the calendar.

◆ Step 1: Data Sync – Full Detail

Objective:

To collect all relevant historical and projected data for the WSSI calculation window, filtering out unnecessary data to reduce processing time and ensure only valid combinations (store, brand, category, warehouse) are considered in the downstream calculations.

17 Date Window:

Derived from the **Retail Week Calendar**:

- Start: `analysis_start_date` of the **first** subperiod
- End: `forecast_end_date` of the **last** subperiod

This ensures the synced data covers:

- LY performance windows
- Current period's actuals
- Future periods for projection

Tables Synced:

Table Name	Type	Purpose
<code>a_sales</code>	System-integrated	Daily sales data (store × SKU × day)
<code>a_keyframe</code>	System-integrated	Daily store stock snapshot
<code>input_wssi_whstock</code>	System-integrated	Warehouse stock snapshot
<code>input_wssi_store_inwards</code>	System-integrated	Store-level stock inflows
<code>input_wssi_wh_inwards</code>	System-integrated	Warehouse-level stock inflows
<code>open_orders</code>	System-integrated	Open purchase orders
<code>git</code>	System-integrated	Goods currently in transit (expected)

Filters Applied:

1. **Enabled Stores** – Only stores marked as active and valid in master store mapping.
2. **Enabled Brands** – Based on brand configuration, filtering is applied before sales/inventory data is pulled.
3. **Enabled Categories** – Filters ensure only relevant product types (e.g., apparel, footwear) are considered.
4. **Enabled Warehouses** – WH data is considered only for warehouses listed in master warehouse list (`a_warehouse` or similar).
5. **Date Filter** – All transactional data must fall within the range defined above.

Sync Output:

Once synced, this data becomes the **foundational layer** for all future aggregations:

- Sales → used for cleaned LY/CY
- Inventory → used for GRN, tagging, opening, inwards
- Inwards → used for calculating Planned Inwards
- WH data → incorporated into projected coverage and OTB

Example:

Let's say the Retail Calendar contains:

Subperiod	Forecast Start	Forecast End	Analysis Start	Analysis End
W01	2025-03-25	2025-03-31	2024-03-26	2024-04-01
W02	2025-04-01	2025-04-07	2024-04-02	2024-04-08
W03	2025-04-08	2025-04-14	2024-04-09	2024-04-15

✓ In this case:

- Start date = 2024-03-26 (analysis start of W01)
- End date = 2025-04-14 (forecast end of W03)

All data synced will lie within this date range only.

◆ Step 2.1: Last Year (LY) Calculation

Objective:

This step focuses on calculating the performance metrics for the same subperiods (weeks) in the **previous year**. These metrics act as a baseline for evaluating how the business is performing this year (CY) compared to last year (LY).

How we define "Last Year":

For each week in the WSSI view (called a **subperiod**), we refer to the **Retail Week Calendar**, which provides two important date ranges:

- `forecast_start_date` and `forecast_end_date` – used for current year
- `analysis_start_date` and `analysis_end_date` – used for last year (LY)

We use the **LY analysis window** to select sales data from last year that aligns with the same week we're analyzing this year.

Data used:

- **Primary source:** `a_sales` table, which contains sales data at a day × store × SKU level.
- **Other mappings used:**
 - SKU to Style → to fetch Brand, Segment, and Fashion Grade
 - Store to Channel → from store master
 - Store to Sales Channel → from `input_wssi_store_sales_channel`

What we calculate:

For each subperiod, we filter the `a_sales` data to include only transactions where the date falls between that subperiod's `analysis_start_date` and `analysis_end_date`.

We then group this filtered data by the following:

- Store
- Channel
- Brand
- Segment
- Fashion Grade
- Sales Channel
- Subperiod

And calculate these key metrics:

Metric	Description	Formula
Net Sales Value	Total revenue excluding discount	Sum of <code>revenue</code>
Unit Sales	Total quantity of units sold	Sum of <code>quantity_sold</code>
Discount Value	Total discount provided	Sum of <code>discount_value</code>
Discount %	Percentage discount offered on top of net sales	<code>discount_value / (revenue + discount_value)</code>
ASP	Average price per unit sold (net of discount)	<code>revenue / quantity_sold</code>

Output:

These calculated values are stored in an intermediate table named `interim_wssi_ly_cleaned_sales`.

 This table stores LY metrics at a store level, which are later aggregated at the format/channel level before being displayed on the WSSI screen.

◆ Step 2.2: Current Year (CY) – Actuals Subperiod Calculation

Objective:

This step calculates the actual sales performance for the **current year** — specifically for those subperiods (weeks) that have already occurred as of the current **run date**. These values reflect what has actually happened so far and are shown as **CY Actuals** on the WSSI screen.

How we identify “Actual” subperiods:

The Retail Week Calendar defines the `forecast_start_date` and `forecast_end_date` for each subperiod.

- We compare these dates with the current **run date** (i.e., the date when the module is being run).
- Any subperiod where the **forecast_end_date is before the run date** is considered an **actual subperiod**.
- Subperiods starting from the run date onwards are considered **projected**, and handled separately (in Step 2.3).

Data used:

- **Primary source:** `a_sales` table (store × SKU × day)
- **Supporting mappings:**
 - SKU to Style (for Brand, Segment, Fashion Grade)
 - Store to Channel (from `a_store`)
 - Store to Sales Channel (from `input_wssi_store_sales_channel`)
 - Retail Week Calendar (for date ranges)

What we calculate:

For each actual subperiod of the current year, we filter the `a_sales` table to include only sales between that subperiod's:

- `forecast_start_date`
- `forecast_end_date`

We group the sales data at the same level as LY:

- Store
- Channel
- Brand
- Segment
- Fashion Grade
- Sales Channel
- Subperiod

Then, we calculate:

Metric	Description	Formula
Net Sales Value	Total revenue from actual sales	Sum of <code>revenue</code>
Unit Sales	Total quantity sold	Sum of <code>quantity_sold</code>
Discount Value	Total discount offered	Sum of <code>discount_value</code>
Discount %	Discount as a percentage of total value	<code>discount_value / (revenue + discount_value)</code>
ASP	Average price per unit sold	<code>revenue / quantity_sold</code>

Output:

The result of this step is stored in a table typically named `interim_wssi_forecast_cleaned_sales` .

 Like LY, this table is also generated at the store level and later aggregated at the format/channel level for display on the WSSI screen.

◆ Step 2.3: Breaking Period AOP into Subperiod-Level

Objective:

AOP (Annual Operating Plan) revenue is typically defined at a **monthly or period level**, but WSSI works at a **weekly level (subperiod)**. This step is solely about **breaking down the period-level AOP revenue into subperiod-level revenue** so that each week has a planned revenue value that can be used for downstream calculations.



Inputs:

- **Source Table:** `performance_parameters`
- Key dimensions:
 - Store
 - Channel
 - Brand
 - Segment
 - Fashion Grade
 - Period (month)
- Key value:
 - `aop_revenue` (at period level)



How the breakdown works:

For each period (e.g., April 2025), the subperiods (weeks) under that period are identified using the **Retail Week Calendar**.

We then determine how to split the `aop_revenue` across those subperiods.

There are two possible scenarios:



Case 1: LY sales exist for the subperiods of the period

1. We fetch LY Net Sales for each subperiod (from `interim_wssi_ly_cleaned_sales`)
2. We calculate each subperiod's **contribution %** to the total LY sales of that period
3. We apply that % to the period AOP to get subperiod AOP

Example:

Period	Subperiod	LY Net Sales	Contribution %	Period AOP = ₹1,00,000	Subperiod AOP
Apr	W01	₹20,000	20%	₹1,00,000	₹20,000
Apr	W02	₹30,000	30%		₹30,000
Apr	W03	₹50,000	50%		₹50,000



Case 2: LY sales = 0 for all subperiods of that period

- If LY data is missing or zero for all subperiods in the period, the `aop_revenue` is split **equally** across all subperiods.

Example:

If April has 3 subperiods and period AOP = ₹90,000 → each subperiod gets ₹30,000

Output:

The result of this step is stored in an intermediate table such as `interim_wssi_subperiod_aop`.

This table contains:

- Subperiod
- Store, Channel, Brand, Segment, Fashion Grade
- Subperiod-level `aop_revenue`

Step 3.1: CY Metrics Calculation (Interim – Store Level)

Objective:

This step prepares the **Current Year (CY)** performance data for each subperiod, distinguishing between **actual** and **projected** weeks based on the run date. The goal is to calculate Net Sales Value, Unit Sales, ASP, Discount%, and Discount Value at the **store level**, which will be aggregated later for the WSSI screen.

17 Identifying Actual vs Projected Weeks:

- From the Retail Week Calendar:
 - If the current **run date** is after the subperiod's `forecast_end_date`, it is marked as an **actual** week
 - Otherwise, the subperiod is treated as **projected**

Data Sources:

Subperiod Type	Source Table
CY Actuals	<code>interim_wssi_forecast_cleaned_sales</code>
CY Projected	<code>interim_wssi_store_fg_subperiod_aop</code>

Metrics and Logic:

For CY Actual Subperiods:

All metrics are directly sourced from the table `interim_wssi_forecast_cleaned_sales`, where calculations are already performed and stored. This includes:

- **Net Sales Value:** Revenue after discount
- **Unit Sales:** Quantity sold
- **ASP:** Revenue ÷ Quantity Sold
- **Discount %:** Discount Value ÷ (Revenue + Discount Value)
- **Discount Value:** Already computed in the table

No further computation is done in this step for actuals.

For CY Projected Subperiods:

These metrics are computed using planning inputs from AOP and `performance_parameters`. Calculations are done as follows:

- **Net Sales Value:** Taken directly from `aop_revenue` for the subperiod

- **Discount %:** Pulled from `performance_parameters`

- **MRP:** Also pulled from `performance_parameters`

- **ASP:**

$$\text{ASP} = \text{MRP} \times (1 - \text{Discount\%})$$

- **Unit Sales:**

$$\text{Unit Sales} = \text{Net Sales} / \text{ASP}$$

- **Discount Value:**

$$\text{Discount Value} = \text{Net Sales Value} \times \text{Discount\%} / 100$$

These values are calculated at the **store level** and later aggregated for WSSI screen display.

Output:

The metrics calculated in this step are not used directly on the screen. They are stored as **interim outputs at store level** in `interim_wssi_act_proj_data`, forming the basis for CY (actual + projected) rows in the WSSI screen after format/channel-level aggregation

◆ Step 3.2: AOP Metrics and COGS Calculation (Interim – Store Level)

Objective:

This step calculates the **AOP metrics** for all weeks in the WSSI calendar and also computes **COGS (Cost of Goods Sold)** based on margin planning. These values are used to populate the AOP row on the WSSI screen and to support downstream financial analysis.

Data Sources:

- `interim_wssi_store_fg_subperiod_aop` → AOP revenue at subperiod level
- `performance_parameters` → MRP, Discount %, and Margin % for each store-channel-brand-segment-fashion_grade combination

Metrics Calculated:

Metric	Formula
Net Sales Value	Taken from <code>interim_wssi_store_fg_subperiod_aop</code>
Discount %	Taken from <code>performance_parameters</code>
MRP	Taken from <code>performance_parameters</code>
ASP	$\text{ASP} = \text{MRP} \times (1 - \text{Discount\%})$
Unit Sales	$\text{Unit Sales} = \text{Net Sales Value} / \text{ASP}$
Discount Value	$\text{Discount Value} = \text{Net Sales Value} \times \text{Discount\%} / 100$
COGS	$\text{COGS} = \text{MRP} \times (1 - \text{Margin\%})$

Output:

- AOP metrics are stored at the **store level** for all subperiods

- **COGS values** are stored in the `interim_wssi_aop_data` table
 - These are later aggregated to the format/channel level
 - average is applied where AOP revenue > 0 to compute final COGS shown on screen

◆ Step 3.3: Aggregating CY, LY, and AOP Metrics for WSSI Screen

🎯 Objective:

All sales and planning metrics calculated so far — including CY (actual + projected), LY, and AOP — have been processed at the **store level**. The purpose of this step is to **aggregate that data to the WSSI screen level**, which operates at the **Format × Brand × Segment × Fashion Grade × Sales Channel × Subperiod** level.

This final aggregation makes the data ready for display on the WSSI user interface.

📊 Source Tables:

Metric Type	Source Table
CY (Actual + Projected)	<code>interim_wssi_act_proj_data</code>
LY	<code>interim_wssi_ly_cleaned_sales</code>
AOP + COGS	<code>interim_wssi_aop_data</code>

📊 Aggregation Dimensions:

The store-level data is aggregated to the following dimensions:

- **Format** (mapped from store codes)
- **Brand**
- **Segment**
- **Fashion Grade**
- **Sales Channel**
- **Subperiod**

🔍 The Format is derived from store metadata using internal mapping logic — typically grouping multiple stores under a planning unit like EBO, MBO, Online, etc.

📊 Metrics Aggregated:

Each of the source tables contributes to the following metrics:

Metric	Aggregation Method
Net Sales Value	Sum of revenue
Unit Sales	Sum of quantity sold
Discount Value	Sum of discount value
ASP	Weighted average = Total Net Sales / Total Units Sold
Discount %	<code>Total Discount Value / (Net Sales + Discount Value)</code>

COGS	Weighted average using Net Sales or AOP revenue as weight: $COGS = \frac{\sum(COGS \times Revenue)}{\sum(Revenue)}$ (only where aop_revenue > 0)
Cover duration	Taken from performance params



Output:

Once aggregated, the data is ready to be displayed on the WSSI screen in the following rows:

- **CY** → Actuals for past weeks, projections for upcoming weeks
- **LY** → Past year's reference performance
- **AOP** → Baseline plan for all weeks

These rows appear under each section:

- Net Sales Value (₹)
- Unit Sales
- ASP
- Discount %
- COGS (only for projected + AOP)



This step completes the backend preparation of all screen-facing rows for sales and revenue performance. Inventory and inwards sections are processed in subsequent steps.

◆ Step 4.1: Style GRN Date Identification



Objective:

Before inventory and sales projections can begin, it's essential to determine **when each style was first received into the retail ecosystem** — this is known as the **GRN (Goods Received Note) date**. This helps determine the starting point of the style's lifecycle and is critical for calculating shelf life, sales channel tagging, and inventory bucketing.



Data Source:

- **a_keyframe** : Daily snapshot of store inventory
 - Contains fields: **date**, **store**, **sku**, **quantity**
 - Each SKU is mapped to a **style** using SKU–style mapping tables



Logic:

1. From **a_keyframe**, identify all the dates on which each style appeared in any store with a **positive quantity**.
2. Take the **minimum date** across all stores — this becomes the **first GRN date** for the style.
3. The GRN is stored at the **style level** (not per store).
4. Result is saved in the table:

interim_wssi_style_first_grn

Columns: `style` , `grn_date`

Example:

Assume the following entries exist in `a_keyframe` :

Date	Store	SKU	Style	Quantity
2024-05-15	STR001	SKU001	STY123	10
2024-05-18	STR002	SKU002	STY123	5
2024-05-20	STR003	SKU003	STY123	8

Although the style **STY123** appears in three different stores on three different dates, the first date where it appears in any store is:

- **GRN Date for STY123** = `2024-05-15`

Even if the style was launched at different times across different stores, we consider the **earliest appearance** system-wide.

Output Table (`interim_wssi_style_first_grn`):

Style	GRN Date
STY123	2024-05-15
STY456	2024-06-01

This GRN date will be used in next step to determine:

- Whether the style is within shelf life
- Whether it should be included in WSSI calculations
- Which sales channel it belongs to (based on inventory flow logic)

◆ Step 4.2: Sales Channel Tagging Based on GRN Date and Inventory Flow

Objective:

Once the **GRN (Goods Received Note) date** of each style has been determined (in Step 4.1), this step decides:

1. Whether the style should be included in WSSI planning
2. If included, what is the **current stage** of the style in its lifecycle
3. What **sales channel** the style should be tagged to for planning its inventory and sales

This classification helps ensure that only relevant inventory is considered, and that it's planned in the correct channel, based on how far the style has progressed through its defined inventory flow.

Inputs Used:

Input Table	Purpose
<code>interim_wssi_style_first_grn</code>	Holds the GRN date per style
<code>input_wssi_inventory_bucket_flow</code>	Defines the flow stages for each inventory bucket
<code>input_wssi_style_inventory_bucket</code>	Links each style to its inventory bucket

Retail Week Calendar	Maps subperiods to dates (forecast start & end)
Run Date	Date on which WSSI is executed
Shelf Life & Lead Time (from inventory flow)	Used to calculate where in the flow the style should be

Core Concept:

Each **inventory bucket** (e.g., X, Y) has a **defined flow** across multiple sales channels (e.g., F1 → F2 → F3). A style that belongs to a bucket progresses through this flow over time, governed by:

- **Shelf Life** (how long it stays in each channel)
- **Lead Time** (transition buffer before moving to the next channel)

The GRN date indicates when the style first entered the system. Based on how much time has passed since then (relative to the run date), we determine the style's current position in the flow — or whether it should be excluded.

Detailed Tagging Logic

Each style goes through one of five tagging cases:

Case 1: GRN Not Found

When this occurs:

- The style has no entry in `a_keyframe`, meaning it hasn't reached any store yet.

Implication:

- This is typically a **future style**, yet to be launched.

Action:

- Assign the style to the **first sales channel** in the inventory bucket flow.
- It is included in WSSI planning as it will potentially contribute to future sales.

Tagging Output

Sales Channel	First channel in flow (e.g., F1)
Description	"GRN not found"

Case 2: GRN Date is Found but Not in Any Subperiod

When this occurs:

- The GRN date exists, but does **not fall into any subperiod** defined in the Retail Calendar.
- This may happen if the GRN is outside the planning window.

Implication:

- The style lies outside the planning horizon and is irrelevant for current cycle.

Action:

- Completely exclude this style from the WSSI module.

Tagging Output

Sales Channel	NA
Description	"GRN date is not in any subperiod"

🟢 Case 3: GRN Date is Found and Falls Within a Valid Subperiod

Once we confirm the GRN falls within the valid WSSI calendar, we check where the style lies in its flow. This breaks down further:

✅ Case 3.1: Within Lifecycle (Resolved)

When this occurs:

- GRN date is valid and recent
- The style is **still within its flow cycle**, based on the number of weeks since GRN

How it's calculated:

- Find the subperiod in which the GRN date lies → call this `starting_subperiod`
- Start calculating shelf life from the **next subperiod**
- Determine current stage in flow by checking how many weeks have passed since GRN

Action:

- Assign the style to the **corresponding channel** as per its flow stage
- Consider it **live** in planning

Tagging Output

Sales Channel	Current stage in bucket flow, (F1/F2)
Description	"Resolved"

✅ Case 3.2: Last Stage

This is a nuanced edge case.

When this occurs:

The following condition must be satisfied:

```
forecast_start_date of run_date_subperiod > forecast_end_date of starting_subperiod
AND
forecast_start_date of run_date_subperiod ≤ GRN date + 15 months
```

Let's break it down:

- `starting_subperiod` = the subperiod where the GRN date falls
- `run_date_subperiod` = subperiod that includes the current run date

Interpretation:

- The run date is already **beyond the GRN's initial subperiod**, so the style has moved past its originally planned stages
- However, the GRN is **still within 15 months**, so the style hasn't completely aged out

Implication:

- The style is **no longer in active flow**, but **still relevant**
- So it is considered to be in its **final channel**

Action:

- Assign the style to the **last channel** in its inventory flow
- Include it in planning, but flag it as aged

Tagging Output

Sales Channel	Last stage in flow, Eg: F3
Description	"Last Stage"

✖ Case 3.3: Aged Out (>15 months old)

When this occurs:

- The GRN date is more than **15 months** before the run date

Implication:

- This is considered **dead stock**
- It is too old to influence current or future planning

Action:

- Completely exclude from WSSI calculations

Tagging Output

Sales Channel	NA
Description	"Remove from inventory"

🔧 Full Example:

Style	GRN Date	Run Date	Days Since GRN	Case	Sales Channel	Description
STY001	—	2025-03-30	—	Case 1	F1	GRN not found
STY002	2023-01-10	2025-03-30	> 780 days	Case 3.3	NA	Remove from Inventory
STY003	2024-06-10	2025-03-30	9 months	Case 3.1	F2	Resolved
STY004	2023-05-15	2025-03-30	22 months	Case 3.3	NA	Remove from Inventory
STY005	2023-12-15	2025-03-30	14.5 months	Case 3.2	F3	Last Stage

◆ Step 4.3: Inventory & Inwards Allocation Across Sales Channels

Objective:

After tagging each style with its correct **sales channel and lifecycle status** (in Step 4.2), this step is about assigning the corresponding **store stock, warehouse stock, and inwards (store + warehouse)** to the right **sales channels** for planning.

This allocation drives the **Inventory Units (U)** and **Inwards Units (U)** rows in the WSSI screen and forms the foundation for downstream metrics like CoGS, OTB, and Weeks of Cover.

Inputs Used:

Input Table	Description
a_keyframe	Store-level stock snapshot (SKU × store × date × qty)
input_wssi_store_inwards	Inwards into stores
input_wssi_whstock	Warehouse stock levels
input_wssi_wh_inwards	Warehouse inwards
input_wssi_style_inventory_bucket	Style → Inventory bucket mapping
input_wssi_inventory_bucket_flow	Inventory bucket → Sales channel flow (F1 → F2 → F3 etc.)
interim_wssi_store_fg_subperiod_aop	AOP revenue broken down to store × subperiod × channel
Step 4.2 Output	Style tagging with assigned sales channel and description

Core Idea:

All four inventory-related inputs (store stock, warehouse stock, store inwards, warehouse inwards) must be **reallocated** from store/warehouse level to a **sales channel**, based on:

1. The style's **current lifecycle stage**
2. Its **assigned sales channel**
3. Whether it's still in its launch phase or has moved beyond
4. How much value it is expected to contribute (via AOP)

Case A: Style is Tagged to the First Sales Channel in its Inventory Flow

Meaning:

- The style is in its **initial launch phase**
- Tagged sales channel matches the **first stage** in its defined flow (e.g., F1 if flow is F1 → F2 → F3)

Action:

- All inventory and inwards are kept **as-is**, based on where they exist
- If stock exists in stores belonging to:
 - F1 → It is counted under F1
 - F2 or F3 → Still retained under those respective channels (physically present, not redistributed)
- Warehouse stock/inwards is assigned as per the AOP contribution(explained in case B)

Allocation Strategy:

Inventory and inwards are handled differently depending on **where they are located**:

Type	Treatment
Store Stock	✅ Retained as-is based on the store's actual sales channel
Store Inwards	✅ Retained as-is based on the receiving store's sales channel
Warehouse Stock	🔄 Split based on AOP contribution across channels (not retained at WH level)
Warehouse Inwards	🔄 Split based on AOP contribution across channels

Why this difference?

- **Store-level stock and inwards** are already physically assigned to specific sales channels through the stores they belong to.
- **Warehouse-level inventory**, however, is still in central holding and needs to be **redistributed** to channels based on where it will likely flow.
 - This is determined by **AOP contribution** of each channel — i.e., how much of the total business for that style is expected from each channel.

✅ Case B: Style is Tagged to a Later Sales Channel (Not First in Flow)

Meaning:

- The style has completed its initial lifecycle(s) and has moved to a **later channel** (e.g., F2 or F3)
- Example: Inventory bucket flow is F1 → F2 → F3 and style is tagged to **F2**

Action:

All inventory and inwards (store + WH) are **reallocated proportionally** based on the **AOP revenue contribution of stores** belonging to the **tagged channel**.

⚠️ Very important:

Only AOP revenue from stores that belong to the tagged sales channel (e.g., F2) is used in the calculation.

Steps:

1. **Collate total stock and inwards** for the style:
 - From all stores and all warehouses
2. **Identify all stores belonging to the tagged sales channel** (e.g., F2)
 - From `input_wssi_store_sales_channel`
3. **From those stores**, get AOP contribution per channel using `interim_wssi_store_fg_subperiod_aop`
4. **Calculate contribution % per channel**:
 - Sum AOP across all stores in that channel
 - Normalize to 100%

5. Redistribute total inventory/inwards in the same ratio

Why:

- The style has aged out of store-specific allocation
- Now inventory is considered a **common pool** to be assigned based on where it can **generate value**

Example:

- Style tagged to **F2**
- Inventory: 1,200 units (store + warehouse)
- Inwards: 800 units
- All F2 stores show below AOP contribution:

Channel	AOP Revenue
Online	₹60,000
EBO	₹30,000
MBO	₹10,000

Split:

- Online: 60%
- EBO: 30%
- MBO: 10%

Final Allocation:

Channel	Inventory Units	Inwards
Online	720	480
EBO	360	240
MBO	120	80

Case C: Style Tagged as 'GRN Not Found'

Meaning:

- The style has not yet been GRNed but inventory exists (e.g., due to pipeline/pre-GRN loading)

Action:

- All inventory and inwards for this style are assigned to the **first channel in its inventory bucket flow**

|  Treated as a new style that will enter the market soon

Why:

- We assume the inventory is part of an upcoming launch and must be planned for its first phase

Example:

- Bucket Flow: F1 → F2 → F3

- Style tagged as 'GRN Not Found'
- WH stock = 1,000
- Store inwards = 600

→ All 1,600 units are assigned to **F1**

✗ Case D: Style is Tagged as 'Remove from Inventory'

🔍 Meaning:

- The style has either:
 - GRN date not in any subperiod
 - GRN is older than 15 months
 - Tagged as aged out

📌 Action:

- **No inventory or inwards are allocated**
- The style is **completely excluded** from planning

🧠 Why:

- It is treated as obsolete
- Planning for this style would distort forecast and stock health metrics

📊 Output Metrics Populated

Metric Name	Formula	Description
Opening Inventory (U)	Store stock + Warehouse stock (reassigned by logic)	Beginning-of-week inventory
Planned Inwards (U)	Store inwards + Warehouse inwards (reassigned)	Expected inward movement for each week

📄 Output Aggregation Level

The final results are rolled up at the following level (to match WSSI screen):

- Format
- Brand
- Segment
- Fashion Grade
- Sales Channel
- Subperiod

UI formulae -

<https://docs.google.com/spreadsheets/d/1NrIfqBWEvuwnp8W06a8TNiBdlhpf26ZYynMfXNkfGoE/edit?gid=1025086892#gid=1025086892>

Question	Answer
What is the purpose of the WSSI module?	Enables week-level sales and inventory planning across formats, providing visibility into LY, CY, and AOP.
What stores are considered for analysis?	The stores that are enabled in the Store config input, and stores that have valid sales channel tagging and AOP defined in performance parameters. Others are excluded.
What is the date range used on which the analysis happen?	All system-integrated inputs are filtered between the earliest <code>analysis_start_date</code> and latest <code>forecast_end_date</code> from the Retail Week Calendar.
How does the Inventory Bucket Flow input help?	It defines the flow of inventory across sales channels — indicating shelf life, lead time, and transitions between channels like F1 → F2 → F3.
What is the purpose of the Style Inventory Bucket input?	Maps each style to an inventory bucket (like X or Y), determining where it launches and how it flows across channels over time.
Why do we need Store Sales Channel Mapping?	Classifies each store into a channel (e.g., F1, F2), which defines inventory movement based on lifecycle and flow logic.
At what level is Performance Parameters data defined?	Store × Brand × Segment × Fashion Grade × Period (month).
Why is the Performance Parameters input important?	Contains AOP revenue, MRP, Discount %, Margin %, and Cover Duration — all used in forward-looking planning.
What are the key columns in the Retail Week Calendar?	<code>subperiod</code> , <code>period</code> , <code>forecast_start_date</code> , <code>forecast_end_date</code> , <code>analysis_start_date</code> , <code>analysis_end_date</code> .
What is the purpose of the Retail Week Calendar input?	Maps time dimension: weeks (subperiods) and months (periods) with associated forecast/analysis dates.
How many types of inputs are used in the WSSI module?	Two: (1) User-defined (uploaded manually), (2) System-integrated (pulled via backend).
How is ASP calculated for future/projected weeks?	$MRP \times (1 - \text{Discount}\%)$ at Store-Brand-Segment-FG-subperiod level, then averaged at the user's screen view level.
What causes a style to be excluded from WSSI?	GRN missing/out of range, aged out, or no valid channel mapping.
Why does Discount % look off for projections?	Because it's based on planning parameters, not historical discounting.
Can OTB or Additional Inwards be edited?	Yes, they are editable for business flexibility.
How does AOP split happen when there is no LY data for period?	AOP is split equally among subperiods if all lack LY data.
How is warehouse inventory allocated?	Based on AOP revenue contribution of the tagged sales channel.
What happens if a style is >15 months old?	It is excluded from planning analysis.
What is the GRN date and why is it important?	The date a style first appeared in any store; used to determine lifecycle stage.
How is COGS calculated?	$MRP \times (1 - \text{Margin}\%)$ from performance parameters, aggregated for planning use.
How is Planned Inwards calculated?	Store Inwards + Warehouse Inwards.

How is Opening Inventory determined?	Store + Warehouse stock, reassigned based on tagging and allocation logic.
What metrics are shown in the Sales Metrics panel?	Net Sales Value, Unit Sales, ASP, Discount %, COGS (for projections).
How is Last Year (LY) ASP calculated?	LY NSV ÷ LY Unit Sales for that subperiod/week.
How is Last Year (LY) Unit Sales calculated?	Aggregated quantity at Format-Brand-Segment-FG-Sales Channel-Subperiod using LY analysis dates.
How is Last Year (LY) Net Sales Value calculated?	Aggregated revenue at Format-Brand-Segment-FG-Sales Channel-Subperiod using LY analysis dates.
At what granularity does WSSI operate?	Subperiod (week) level across Format → Channel → Brand → Segment → Fashion Grade → Sales Channel.